

## Polynomials in Number Theory — Solutions

**Problem 1.** (International Mathematical Excellence Olympiad 2019 Problem 4) Call a two-element subset of  $\mathbb{N}$  *cute* if it contains exactly one prime number and one composite number. Determine all polynomials  $f \in \mathbb{Z}[x]$  such that for every cute subset  $\{p, q\}$ , the subset  $\{f(p) + q, f(q) + p\}$  is cute as well.

**Solution.** The only polynomial  $f$  is the zero polynomial. It is clear that the zero polynomial satisfies the condition.

If  $f(x) = c$  is a constant polynomial, then  $\{c + p, c + q\}$  is cute whenever  $\{p, q\}$  is cute. Suppose there exists a prime  $p$  such that  $c + p$  is a prime. Then for any composite number  $m$ , the integer  $c + m$  is composite. Thus, by considering the cute set  $\{c + p, m\}$ , we know that  $2c + p$  is a prime. Inductively, we find that  $kc + p$  is a prime for any positive integer  $k$ . In particular,  $pc + p = p(c + 1)$  is a prime. This only holds when  $c = 0$ .

Next, suppose there exists a prime  $p$  such that  $c + p$  is composite. Then for any composite number  $m$ , the integer  $c + m$  is a prime. But we can easily find infinitely many composite numbers  $m$  having the same parity as  $c$ . This implies  $c + m$  is even, but  $c + m > 2$  since  $m$  can be arbitrarily large, contradiction.

It remains to consider non-constant polynomials. Since  $f$  is a polynomial, the parity of  $f(n)$  is the same as that of  $f(1)$  if  $n$  is odd.

- If  $f(n)$  is odd for any odd  $n$ , we consider the cute set  $\{3, 3m\}$  where  $m \geq 3$  is odd. This shows  $\{f(3) + 3m, f(3m) + 3\}$  is cute. Note that both  $f(3) + 3m$  and  $f(3m) + 3$  are even, and so one of them is 2. For sufficiently large  $m$ , we have  $f(3) + 3m > 2$ . Therefore, we must have  $f(3m) + 3 = 2$ . This shows  $f$  is a constant polynomial, contradiction.
- If  $f(n)$  is even for any odd  $n$ , we consider the cute set  $\{2, 3m\}$  where  $m \geq 3$  is odd. This shows  $\{f(2) + 3m, f(3m) + 2\}$  is cute. As above, only finitely many  $m$  may satisfy  $f(3m) + 2 = 2$ . As  $f(3m) + 2$  is even,  $f(2) + 3m$  is a prime for all sufficiently large odd  $m \geq 3$ . This is clearly impossible (for example, by considering  $m \equiv -3^{-1}f(2) \pmod{7}$ ).

Therefore,  $f$  can only be the zero polynomial. □

**Problem 2.** (Canada MO 2016 Problem 3) Find all polynomials  $P(x)$  with integer coefficients such that  $P(P(n) + n)$  is a prime number for infinitely many integers  $n$ .

**Solution.** The solutions are  $P(x) = -2x + c$  for some odd integer  $c$ , or  $P(x) = p$  for some prime  $p$ .

Since  $P(n) + n \equiv n \pmod{P(n)}$  and  $P$  is a polynomial, we have

$$P(P(n) + n) \equiv P(n) \equiv 0 \pmod{P(n)}.$$

As  $P(P(n) + n)$  is a prime, we have  $P(n) = \pm 1$  or  $P(P(n) + n) = \pm P(n)$ .

If  $P(n) = \pm 1$  for infinitely many  $n$ , then  $P$  must be the constant polynomials  $P(x) = \pm 1$ . But then  $P(P(n) + n) = \pm 1$  is never a prime number.

If  $P(P(n) + n) = \pm P(n)$  for infinitely many  $n$ , then  $P(P(n) + n) = P(n)$  or  $P(P(n) + n) = -P(n)$  holds for all  $n$ . Let  $\deg P = d$ . If  $d \geq 2$ , by comparing the degrees on both sides, we get the contradiction  $d^2 = d$ . Thus,  $d \leq 1$ .

Let  $P(x) = ax + b$  for some constants  $a, b$ . Then we have

$$P(P(n) + n) = P(an + b + n) = a(a + 1)n + ab + b = (a + 1)(an + b).$$

This is equal to  $\pm P(n) = \pm(an + b)$  if and only if  $a = 0$  or  $a = -2$ .

If  $a = 0$ ,  $P(x) = b$  is a constant polynomial. Clearly,  $b$  must be equal to some prime number  $p$ . In that case the condition holds.

If  $a = -2$ ,  $P(x) = -2x + b$  for some integer  $b$ . Note that  $P(P(n) + n) = 2n - b$ . If  $b$  is even,  $P(P(n) + n)$  is even, and hence it is not a prime except  $n = \frac{b+2}{2}$ . The condition does not hold. If  $b$  is odd,  $P(P(n) + n)$  takes up all odd values, and so it is a prime for infinitely many  $n$ .  $\square$

**Problem 3.** (ELMO Shortlist 2018 N1) Determine all nonempty finite sets of positive integers  $\{a_1, \dots, a_n\}$  such that  $a_1 \cdots a_n$  divides  $(x + a_1) \cdots (x + a_n)$  for every positive integer  $x$ .

**Solution.** Any set  $\{1, 2, \dots, n\}$  satisfies the condition.

Let  $P(x) = (x + a_1) \cdots (x + a_n)$  be a monic polynomial of degree  $n$ . Since  $a_1 \cdots a_n$  divides  $P(x)$ , it divides all finite differences of  $P(x)$ . (The finite differences are defined as follows. Let  $\Delta^1 P(x) = P(x + 1) - P(x)$ , and let  $\Delta^{k+1} P(x) = \Delta^k P(x + 1) - \Delta^k P(x)$  for each  $k \in \mathbb{Z}^+$ . It can be proved that if  $\deg P = n$ , then  $\Delta^n P(x)$  is  $n!$  times the leading coefficient of  $P(x)$ .)

In particular, we have  $a_1 \cdots a_n \mid \Delta^n P(x) = n!$ . But since  $a_1, \dots, a_n$  are distinct positive integers, their product is at least  $n!$ . Therefore, equality holds and we must have  $\{a_1, \dots, a_n\} = \{1, 2, \dots, n\}$ .

Since  $\frac{(x + 1) \cdots (x + n)}{n!} = \binom{x + n}{n}$  is an integer, we have  $n! \mid (x + 1) \cdots (x + n)$ . The condition is satisfied.  $\square$

**Problem 4.** (ELMO 2016 Problem 4) Big Bird has a polynomial  $P$  with integer coefficients such that  $n$  divides  $P(2^n)$  for every positive integer  $n$ . Prove that Big Bird's polynomial must be the zero polynomial.

**Solution.** Let  $p$  be an arbitrary odd prime, and let  $m$  be any positive integer. By the Fermat little theorem, we have  $2^{mp} \equiv 2^m \pmod{p}$ . It follows that  $p \mid P(2^{mp}) - P(2^m)$ . Since it is given that  $mp \mid P(2^{mp})$ , and clearly  $p \mid mp$ , we deduce  $p \mid P(2^m)$ . As this holds for all odd primes  $p$ , we must have  $P(2^m) = 0$ . Now, since  $P(n) = 0$  for infinitely many  $n$ 's, the Big Bird's polynomial must be the zero polynomial.  $\square$

**Problem 5.** (Poland MO Finals 2003 Problem 3) Find all polynomials  $W$  with integer coefficients satisfying the following condition: For every natural number  $n$ ,  $2^n - 1$  is divisible by  $W(n)$ .

**Solution.** The only such polynomials are  $W(n) = 1$  and  $W(n) = -1$ .

Since  $W(1) \mid 2^1 - 1 = 1$ , the only possible constant polynomials are  $W(n) = 1$  and  $W(n) = -1$ . Obviously, both satisfy the conditions. We now assume  $W(n)$  is not a constant polynomial. Then there exists a prime  $p$  such that  $p \mid W(m)$  for some integer  $m$ . Since  $p \mid W(m) \mid 2^m - 1$ ,  $p$  must be odd.

By the Chinese remainder theorem, since  $(p, p-1) = 1$ , there exists a positive integer  $n$  such that

$$\begin{cases} n \equiv m \pmod{p}, \\ n \equiv 1 \pmod{p-1}. \end{cases}$$

For this  $n$ , we have  $p \mid W(n) - W(m)$ , and hence  $p \mid W(n)$ , and also  $2^n \equiv 2 \pmod{p}$  by the Fermat little theorem. By the given condition, we have  $p \mid W(n) \mid 2^n - 1$ , while we have shown that  $2^n - 1 \equiv 1 \pmod{p}$ . This contradicts  $p > 1$ . This completes the proof.  $\square$

**Problem 6.** Find all polynomials  $P(x)$  with integer coefficients such that  $P(n)$  divides  $n^{n-1} - 1$  for all sufficiently large positive integer  $n$ .

**Solution.** The answers are  $P(x) = \pm 1$ ,  $P(x) = \pm(x-1)$  and  $P(x) = \pm(x-1)^2$ .

For each prime number  $p$ , suppose  $p \mid P(m)$  for some sufficiently large positive integer  $m$ . This implies  $p \mid m^{m-1} - 1$ . Note that this shows  $(p, m) = 1$ . By the Chinese remainder theorem, since  $(p, p-1) = 1$ , there exists a sufficiently large positive integer  $n$  such that

$$\begin{cases} n \equiv m \pmod{p}, \\ n \equiv m+1 \pmod{p-1}. \end{cases}$$

For this  $n$ , we have  $p \mid P(n) - P(m)$ , and hence  $p \mid P(n) \mid n^{n-1} - 1$ , and also

$$n^{n-1} - 1 \equiv m^{n-1} - 1 \equiv m^m - 1 \pmod{p}$$

by the Fermat little theorem. Since we also have  $m^{m-1} \equiv 1 \pmod{p}$ , we get

$$m - 1 \equiv m \cdot m^{m-1} - 1 = m^m - 1 \equiv 0 \pmod{p}.$$

In other words, whenever  $p \mid P(m)$  for large  $m$ , we must have  $p \mid m - 1$ .

Now, for each large prime  $p$ , since the only prime divisor of  $(p+1) - 1 = p$  is  $p$ , the only possible prime divisor of  $P(p+1)$  is  $p$  from above. Also, for sufficiently large odd prime  $p$ , we have

$$P(p+1) \mid (p+1)^p - 1 = \sum_{r=0}^p \binom{p}{r} p^r - 1 = p^2 + \sum_{r=2}^p \binom{p}{2} p^r.$$

Note that  $p^3 \mid \binom{p}{2} p^r$  for  $r \geq 2$ . Therefore,  $P(p+1)$  can only be  $\pm 1$ ,  $\pm p$  and  $\pm p^2$ . Thus, there are infinitely many  $p+1$  such that  $P(p+1)$  has the same formula. As  $P(x)$  is a polynomial, the same formula holds for all  $x$ . Thus,  $P(x)$  can only be one of  $P(x) = \pm 1$ ,  $P(x) = \pm(x-1)$  and  $P(x) = \pm(x-1)^2$ .

It remains to check that all these are solutions. Indeed, it suffices to prove  $(n-1)^2 \mid n^{n-1} - 1$  for all integers  $n > 1$ . Using the binomial theorem, we have

$$n^{n-1} - 1 = ((n-1) + 1)^{n-1} - 1 \equiv 1 + \binom{n-1}{1}(n-1) - 1 \equiv 0 \pmod{(n-1)^2}.$$

So we are done. □

**Problem 7.** (RMM Shortlist 2018 N1) Determine all polynomials  $f$  with integer coefficients such that  $f(p)$  is a divisor of  $2^p - 2$  for every odd prime  $p$ .

**Solution.** The answers are  $f(x) = \pm 1, \pm 2, \pm 3, \pm 6, \pm x, \pm 2x$ .

Let  $f(x) = x^k g(x)$  such that  $x$  does not divide  $g(x)$ . Since  $f(3) \mid 6$ , we must have  $k = 0, 1$ . Note that  $g(p) \mid f(p) \mid 2^p - 2$ . For each prime  $p$ , suppose  $q$  is a prime different from  $2, 3, p$  such that  $q \mid g(p)$ . Since  $(q, q-1) = (p, q) = (-1, q-1) = 1$ , by the Chinese remainder theorem and Dirichlet's theorem, there exists a prime  $p'$  such that

$$\begin{cases} p' \equiv p \pmod{q}, \\ p' \equiv -1 \pmod{q-1}. \end{cases}$$

For this prime  $p'$ , we have  $q \mid g(p') - g(p)$ , and hence  $q \mid g(p') \mid 2^{p'} - 2$ . Also, we have

$$0 \equiv 2(2^{p'} - 2) = 2^{p'+1} - 4 \equiv 1 - 4 = -3 \pmod{q}$$

by the Fermat little theorem. This is impossible since  $q \neq 3$ .

Thus, we have shown that the only possible prime divisors of  $g(p)$  are  $2, 3, p$ . As  $x$  does not divide  $g(x)$ , only finitely many primes  $p$  may satisfy  $p \mid g(p)$  (which are those primes dividing the constant term of  $g$ ). Since  $g(p) \mid 2^p - 2$  and  $2^p - 2 \equiv 2 \pmod{4}$ , we can only have  $v_2(g(p)) = 0, 1$ . Also, for  $p \equiv 5 \pmod{6}$ , we have

$$2^p - 2 \equiv 2^5 - 2 = 30 \equiv 3 \pmod{9},$$

and so  $v_3(g(p)) = 0, 1$ . Therefore, there are infinitely many primes  $p$  such that  $g(p)$  is one of  $\pm 1, \pm 2, \pm 3, \pm 6$ . So  $g(x)$  is also one of these. It follows that  $f(x)$  is one of  $\pm 1, \pm 2, \pm 3, \pm 6, \pm x, \pm 2x, \pm 3x, \pm 6x$ .

For each odd prime  $p$ , it is clear that  $2 \mid 2^p - 2$  and  $3 \mid 2^p - 2$ . Also, we have  $p \mid 2^p - 2$  by the Fermat little theorem. Therefore, we have  $6p \mid 2^p - 2$  possibly except  $p = 3$ . We check that  $2^3 - 2 = 6$ , which is only divisible by  $6 = 2(3)$  but not  $3(3) = 9$  and  $6(3) = 18$ . Therefore,  $f(x) = \pm 3x, \pm 6x$  are rejected, while all others are solutions. □

**Problem 8.** (China TST 2021 Test 1 Problem 4) Let  $f(x), g(x)$  be two polynomials with integer coefficients. It is known that for infinitely many primes  $p$ , there exists an integer  $m_p$  such that

$$f(a) \equiv g(a + m_p) \pmod{p}$$

holds for all  $a \in \mathbb{Z}$ . Prove that there exists a rational number  $r$  such that

$$f(x) = g(x + r).$$

**Solution.** We first prove the following claim (which is a standard result).

**Claim.** Let  $h$  be a nonzero polynomial of degree  $d$  with integer coefficients, and let  $p$  be a prime greater than  $d$  such that  $p \mid h(a)$  for all integers  $a$ . Then  $p$  divides all coefficients of  $h$ .

*Proof.* Considering modulo  $p$ , we may assume the leading coefficient  $c$  of  $h$  is not divisible by  $p$  (or otherwise the claim is proved). Since  $p$  divides  $h(a)$  for all integers  $a$ , it also divides all finite differences  $\Delta^k h(a)$ . In particular, we have  $p \mid \Delta^d h(a) = d! \cdot c$ . This is a contradiction as  $p > d$  and  $p \nmid c$ .  $\square$

In the following,  $p$  always refer to a prime  $p$  satisfying the given condition. Applying the claim to those sufficiently large  $p$ , we know that  $p$  divides all coefficients of  $f(x) - g(x + m_p)$ . Let

$$\begin{aligned} f(x) &= c_n x^n + c_{n-1} x^{n-1} + \cdots + c_0, \\ g(x) &= d_n x^n + d_{n-1} x^{n-1} + \cdots + d_0 \end{aligned}$$

where at least one of  $c_n, d_n$  is nonzero (if both  $f \equiv g \equiv 0$ , the result holds trivially). If  $d_n = 0$ , then  $p \mid c_n$  for infinitely many primes  $p$  from above, contradicting  $c_n \neq 0$ . This proves  $d_n \neq 0$ . Now, we have

$$\begin{aligned} f(x) - g(x + m_p) &= \sum_{j=0}^n c_j x^j - \sum_{i=0}^n d_i (x + m_p)^i \\ &= \sum_{j=0}^n c_j x^j - \sum_{i=0}^n d_i \sum_{j=0}^i \binom{i}{j} m_p^{i-j} x^j \\ &= \sum_{j=0}^n c_j x^j - \sum_{j=0}^n \sum_{i=j}^n d_i \binom{i}{j} m_p^{i-j} x^j \\ &= \sum_{j=0}^n \left( c_j - \sum_{i=j}^n d_i \binom{i}{j} m_p^{i-j} \right) x^j. \end{aligned}$$

If  $n = 0$ ,  $p \mid c_0 - d_0$  for all large  $p$ , and hence  $f \equiv g$ . The result holds trivially. If  $n \geq 1$ , the coefficient  $c_{n-1} - d_{n-1} - d_n n m_p$  of  $x^{n-1}$  is divisible by  $p$ . Therefore, we have

$$m_p \equiv (c_{n-1} - d_{n-1})(d_n n)^{-1} \pmod{p}.$$

In view of this, we can take  $r = \frac{c_{n-1} - d_{n-1}}{d_n n} \in \mathbb{Q}$ . Then we have  $m_p \equiv r \pmod{p}$  for infinitely many primes  $p$ , and hence

$$f(a) \equiv g(a + m_p) \equiv g(a + r) \pmod{p}$$

for all integers  $a$  for infinitely many primes  $p$ . Thus,  $f(x) = g(x + r)$ .  $\square$